

ITA0606 – MACHINE LEARNING

EXP 7

PROGRAM:

Program to implement Logistic Regression

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

Step 1: Load Iris dataset

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

Convert into binary classification

0 = Setosa, 1 = Not Setosa

```
y = (y == 0).astype(int)
```

Step 2: Split dataset

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.3,
```

```
    random_state=42
```

```
)
```

Step 3: Create Logistic Regression model

```
model = LogisticRegression()
```

```
# Step 4: Train the model
```

```
model.fit(X_train, y_train)
```

```
# Step 5: Predict test data
```

```
y_pred = model.predict(X_test)
```

```
# Step 6: Display actual and predicted values
```

```
print("Actual Values:")
```

```
print(y_test)
```

```
print("\nPredicted Values:")
```

```
print(y_pred)
```

```
# Step 7: Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:")
```

```
print(accuracy * 100, "%")
```

```
# Step 8: Display confusion matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

```
# Step 9: Predict a new sample
```

```
new_flower = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(new_flower)
```

```
print("\nNew Flower Prediction:")
```

```
if prediction[0] == 1:
```

```
    print("Setosa")
```

```
else:
```

```
    print("Not Setosa")
```

OUTPUT:

```
... Actual Values:
[0 1 0 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 1 1 1 0 1 1 0 0
 1 1 1 0 0 0 1 1]

Predicted Values:
[0 1 0 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 1 1 1 0 1 1 0 0
 1 1 1 0 0 0 1 1]

Accuracy:
100.0 %

Confusion Matrix:
[[26  0]
 [ 0 19]]

New Flower Prediction:
Setosa
```